Types of Operating Systems

Task – 1 : Create a table comparing at least three types of operating systems (such as Windows, Linux, and Android) based on their main features, target devices, and examples of apps you use on each.

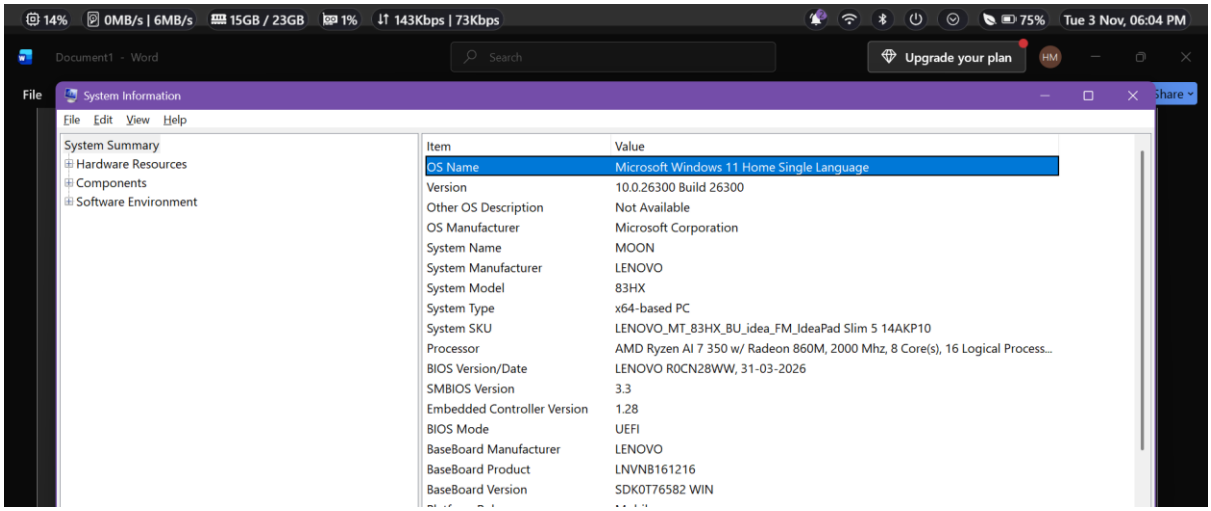
Solution : An Operating System (OS) is system software that manages computer hardware, software applications, memory, files, and other resources. It provides an interface between the user and the device.

Operating System	**Main Features**	**Target Devices**	**Examples of Apps**
Windows	User-friendly graphical interface, multitasking, wide software compatibility, and gaming support.	Laptops and desktop computers.	Google Chrome, MS Word, Instagram, VLC Media Player.
Linux	Open-source, highly customizable, secure, stable, and supports command-line operations.	Computers, servers, and embedded systems.	Firefox, LibreOffice, VLC, Terminal.
Android	Touchscreen interface, app permissions, notifications, widgets, and Google Play Store support.	Smartphones, tablets, and smart TVs.	WhatsApp, Instagram, Zomato, Paytm.

Task – 2 : Identify the operating system running on your phone and your laptop/desktop. Write down two unique features of each OS that affect your daily use of popular apps like Instagram or Zomato.

Solution :

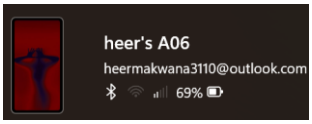
- Laptop's OS



Two important features:

1. Multitasking: Windows 11 supports Snap Layouts and multiple windows, allowing me to use applications such as Instagram in a browser while working on documents.
2. Windows Security: Built-in security features such as Microsoft Defender help protect the system against malware and suspicious applications while browsing websites like Zomato.

- Mobile's OS



Two important features:

1. App Permissions: Android allows users to control permissions such as camera, microphone, contacts, and location. For example, Instagram needs camera access to capture photos and videos.
2. Notifications and Background Apps: Android allows apps like WhatsApp to send notifications and receive messages even when the app is not actively open.

Task – 3 : List three security risks that are specific to mobile operating systems like Android or iOS, and explain how these risks could impact apps you use (for example, WhatsApp or Paytm).

Solution : Mobile operating systems face various security threats because smartphones store personal information, financial details, and private communications.

Security Risk	**Explanation**	**Impact on Apps**
1. Malicious Applications	Fake or infected applications may contain malware that steals personal information or damages the device.	A fake WhatsApp application could steal login credentials or private messages.
2. Excessive App Permissions	Some applications request unnecessary access to contacts, camera, microphone, or location.	A malicious application could access sensitive information or misuse personal data associated with Paytm.
3. Unpatched OS Vulnerabilities	Outdated mobile operating systems may contain security vulnerabilities that attackers can exploit.	Attackers could exploit vulnerabilities to access sensitive information, compromise WhatsApp accounts, or put Paytm data at risk.

Task – 4 : Research and describe how a cloud-based operating system (such as Chrome OS or a virtual desktop environment) is different from a traditional desktop OS. Give one example of how this difference changes the way you access apps like YouTube or Flipkart.

Solution : A cloud-based operating system relies heavily on internet services, cloud storage, and web-based applications. It allows users to access applications and files through the internet.

A traditional desktop operating system installs and manages applications directly on the computer and uses local hardware and storage to run most programs.

Difference between Cloud-Based OS and Traditional Desktop OS

Basis	**Cloud-Based OS (ChromeOS)**	**Traditional Desktop OS (Windows 11)**
Application Access	Mainly through web applications and online services.	Through installed applications and web browsers.
Storage	Uses cloud storage such as Google Drive.	Uses local storage such as HDD or SSD.
Internet Requirement	Internet is important for most cloud-based activities.	Many applications work without an internet connection.
Updates	Updates are generally managed automatically.	Updates are managed through Windows Update.
Device Dependency	Files and settings can be synchronized across devices.	Applications and files are generally stored on the individual computer.